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ORIGINAL PAPER

Assessing the environmental costs of a thermal power plant in tehran using life cycle analysis

N Javidi[1] · G Nabi Bidhendi[1] · O Tavakoli[2] · N Mehrdadi[1]

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Abstract

This study assesses the environmental and economic impacts of a natural gas-fired thermal power plant in Tehran using life cycle analysis and environmental cost accounting. The analysis considers key environmental categories, including global warming potential, acidification potential, eutrophication potential, water consumption, thermal pollution, and particulate matter formation. This study used a life cycle analysis framework integrated with environmental cost accounting to assess the environmental costs of a natural gas-fired thermal power plant in Tehran. The study covered the entire life cycle of the power plant, including fuel extraction, transportation, electricity generation, greenhouse gas emissions, and waste management. The environmental impacts were then quantified in financial terms to estimate the economic burden associated with these impacts, providing a comprehensive view of the plant's environmental footprint. The results show that the plant emits 451.5 kg CO_2 equivalent per megawatt-hour, with a total annual emission of 1.63 million tons of CO_2 equivalent, which significantly contributes to global climate change. The plant's high levels of NOx and SOx emissions, which contribute to acidification and air pollution, are also of concern, along with the plant's annual water consumption of 10.8 million cubic meters, which leads to water shortages and thermal pollution in Tehran. The economic costs associated with these environmental impacts are estimated at $31.20 per megawatt-hour, indicating hidden social costs related to public health and environmental degradation. The study emphasizes the need for policy interventions, such as upgrading emission control technologies, reducing water consumption through dry cooling systems, and transitioning to renewable energy sources to reduce the environmental and economic burdens of thermal power generation. This study also emphasizes the importance of understanding the water-energy nexus in managing resource interdependencies and achieving long-term sustainability in urban energy systems.

Keywords Environmental impacts · Water consumption · Environmental cost accounting · Global warming potential · CO_2 emissions

Introduction

The rapid expansion of urban populations and industrial activities has amplified the global demand for energy, with thermal power plants playing a significant role in meeting this demand. However, the environmental implications of these plants, particularly in urban areas, are substantial and warrant comprehensive evaluation. Tehran, the capital city of Iran, faces pressing environmental challenges due to its dependency on thermal power plants for electricity generation. These plants, which primarily use natural gas as fuel, contribute to air pollution, greenhouse gas (GHG) emissions, and other environmental stresses that threaten both public health and the sustainability of urban ecosystems (IEA, 2018). Given Tehran's geographical and population density constraints, understanding the full extent of these environmental impacts is crucial for informing energy policy and sustainable urban planning.

The environmental impacts of thermal power plants encompass a wide range of pollutants. In particular, the combustion of natural gas releases significant amounts of carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O), all of which are potent GHGs that contribute to global warming (Piriyeva et al. 2024). Additionally, the emission of sulfur

Editorial responsibility: Samareh Mirkia.

✉ O Tavakoli
otavakoli@ut.ac.ir

[1] Faculty of Environment, University of Tehran, Tehran, Iran

[2] School of Chemical Engineering, College of Engineering, University of Tehran, Tehran 14176, Iran

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oxides (SOx), nitrogen oxides (NOx), and particulate matter (PM) exacerbates air quality problems, leading to respiratory diseases and other health complications in densely populated urban centers like Tehran (Kheirbek et al. 2014). Moreover, thermal power plants require substantial amounts of water for cooling, which, when discharged as heated wastewater, can disrupt local aquatic ecosystems. These multifaceted environmental challenges highlight the need for a systematic and holistic evaluation of the environmental costs associated with the operation of thermal power plants.

Life Cycle Analysis (LCA) offers a robust framework for assessing the environmental impacts of energy production systems across their entire life cycle, from raw material extraction to waste disposal (Smith et al. 2021). By adopting a cradle-to-grave approach, LCA facilitates the quantification of environmental burdens at every stage of a thermal power plant's operational life. This method allows for the identification of key environmental hotspots, such as fuel extraction, combustion, and emission control, as well as the associated economic costs of these environmental impacts. As LCA integrates both direct and indirect environmental effects, it provides a comprehensive picture that is essential for informed decision-making in energy policy and environmental management (Wang & Tang 2022).

Previous studies have applied LCA to various energy systems, including coal-fired, gas-fired, and renewable energy sources, to understand their environmental costs. For example, N.M. Piriyeva and colleagues also published a study in 2024 titled Engineering Approaches to Minimizing the Environmental Impact of Thermal Power Plants and found that CO_2 emissions are the most important factor in global warming potential (Piriyeva et al. 2024). With this in mind, Yousefi et al. conducted a 2023 study on life cycle costing and joint life cycle costing to address the energy, cost, and midpoint/endpoint environmental impacts of Tehran's electricity generation/supply industry from start to finish. According to the ReCiPe model, photovoltaic and biogas combined heat and power (CHP) technologies have the highest ecological toxicity, eutrophication, human toxicity, ionizing radiation, and mineral resource depletion. In comparison, fossil technologies, especially gas turbines, have the highest global warming impacts and the formation of specific substances and ozone, and end-point damages to ecosystems, human health, and resources. According to the cumulative energy demand method, the gas turbine has the highest energy demand (16.08 MJ/kWh), while biogas-CHP has the lowest (1.38 MJ/kWh). In contrast, biogas-CHP has the highest cost of electricity supply (US$1.07/kWh), while hydropower has the lowest cost (US$0.39/kWh) and midpoint/point-zero environmental impacts (Yousefi et al. 2023).

Life Cycle Analysis (LCA) is a technique that has been used in many studies before, but in this study, it was decided to use LCA from a new perspective. Where numbers are used to present a report, it will both be more documented and attract more attention. The study attempts to report environmental impacts such as greenhouse gas emissions, air pollution, water consumption and resource depletion in financial terms using environmental cost accounting (ECA) methods to provide a clearer understanding of the economic burden of plant operations. Combining LCA and ECA allows for a comprehensive assessment that not only identifies environmental hot spots, but also assesses the wider economic consequences of continued reliance on thermal energy in urban areas.

By connecting environmental impacts to their economic costs, this study provides a valuable framework for policymakers and urban planners to consider when designing energy strategies for Tehran. Furthermore, the findings of this research can serve as a model for other urban areas facing similar challenges from thermal power generation. The study aims to contribute to the global discourse on sustainable energy practices by demonstrating the importance of integrating environmental and economic assessments in energy planning.

Literature review

The environmental impacts of thermal power plants have been a topic of extensive study in the context of sustainability and energy policy. Life Cycle Analysis (LCA) has emerged as one of the most robust methodologies for evaluating the environmental costs associated with energy generation systems, offering a comprehensive approach by considering the entire life cycle of a product or system from raw material extraction to disposal (Smith et al. 2021).

Life cycle analysis in energy systems

LCA has been widely used to assess the environmental impact of various energy systems, from fossil fuels to renewable sources. In the context of natural gas-fired thermal power plants, LCA studies typically focus on greenhouse gas emissions, air pollutants, water usage, and resource depletion across different life cycle stages, including fuel extraction, transportation, combustion, and waste management (Piriyeva et al. 2024). One of the earliest comprehensive studies by Boustead and Hancock (1979) (Boustead & Hancock 1979) provided a detailed LCA of conventional power plants, setting a foundation for further research. Since then, multiple studies have built on this framework to quantify the specific environmental impacts of energy production technologies.

For example, Delphine Ramon and colleagues conducted an LCA for a Greek natural gas combined cycle power plant in 2023 (Ramon et al. 2023), highlighting that the majority of environmental impacts come from the combustion phase, which accounts for over 60% of total CO_2 emissions. The study found that although natural gas is often touted as a cleaner alternative to coal, it still produces significant amounts of greenhouse gases and other pollutants over its entire life cycle. Similarly, other researchers have analyzed natural gas power plants in the United States and found that methane leaks during extraction and transportation make a significant contribution to the overall environmental footprint of these plants. These findings are particularly relevant in the context of Tehran, where natural gas is the main fuel source for electricity generation.

In a study conducted by Yousefi et al., it was found that the electricity industry's damages to ecosystems (1.99E-09 species.year/kWh) and human health (7.71E-07 DALY/kWh) are mainly caused by the global warming effects of power plants (0.603 kg-CO2/kWh), while the damages to resources (0.101 USD/kWh) are primarily caused by their high non-renewable energy demand (12.36 MJ/kWh). Transmission/distribution networks, despite their small contribution to energy/environmental impacts, account for 48.3% of the total electricity supply costs (1.44 USD/kWh). The findings indicate that replacing diesel from Tehran's electricity mix can significantly increase the sustainability of electricity supply. In line with this increase, the decreasing trend of air pollution should also be considered (Yousefi et al. 2023). Unfortunately, when it comes to electricity supply, we only think about providing the power we need and are completely oblivious to numerous parameters that can increase the rate of air pollution.

Environmental costs of thermal power plants

Studies by Cui et al. in China (2021) and Shahbaz et al. (2020) in BRICS countries show a strong correlation between proximity to thermal power plants and increased rates of respiratory diseases, especially in children and the elderly.

In addition, water consumption is another significant environmental cost of thermal power plants. A study by Seema A. Nihalani et al. (Nihalani & Mishra 2020) noted the high water consumption in power plant cooling systems and noted that in water-scarce areas such as Tehran, this can lead to competition between industrial and domestic water needs. Water consumption is particularly high in natural gas combined cycle (NGCC) power plants, where both power generation and cooling require significant volumes of water. In Tehran, water consumption by thermal power plants further strains already limited water resources, contributing to ecosystem disruption and reduced water availability for agriculture and human consumption.

While all energy sources have inherent trade-offs, understanding the differences and impacts of different energies is crucial for optimizing the energy portfolio. As such, the advice for policymakers and industry leaders is clear. Coordinated actions are needed to refine these energy solutions, focusing on reducing environmental impacts while balancing economic and social benefits. Such strategic guidance requires governments involved in the energy supply sectors to improve resource production and energy production performance so that they can collectively make the best possible contribution to a more sustainable energy mix. This is essential to mitigate the adverse effects of climate change and ensure the sustainability of the energy system (Al Mubarak et al. 2024).

Economic valuation of environmental impacts

Several studies have attempted to quantify the economic costs associated with the environmental impacts of thermal power generation. The Environmental Cost Accounting (ECA) approach is widely used to assign monetary values to environmental impacts, such as the social cost of carbon (SCC) for GHG emissions and health costs for air pollution. For instance, the Stern Review (2006) estimated the global social cost of carbon at $85 per ton of CO_2, though more recent estimates by the U.S. Estimates of the global economic losses from carbon dioxide (CO_2) emissions can inform climate policy. The social cost of carbon (SCC) quantifies these losses by specifying how today's additional CO_2 emissions will affect future economic outcomes through climate change. Previous estimates have shown that large, warming-induced increases in energy costs can dominate the SCC7,8, but they rely on models that are spatially ambiguous and not closely related to the data. In particular, different researchers have predicted that emitting one ton of CO_2 today would reduce total future energy costs, and most estimates have been valued at between US$3 and US$1, depending on discount rates. The results of this research are based on an architecture that integrates global data, econometrics, and climate science to estimate local losses around the world. Notably, we predict that emerging economies in the tropics will increase electricity consumption significantly due to warming, requiring critical infrastructure planning. However, the reduction in warming in colder countries will offset this increase globally. We estimate that annual global electricity consumption in 2099 will increase by about 4.5 exajoules (7% of current global consumption) for every 1 °C increase in global mean surface temperature (GMST), while direct consumption of other fuels will decrease by about 11.3 exajoules (7% of current global consumption) for every 1 °C increase in

GMST. Overall, global data suggest that many populations will remain too poor for much of the twenty-first century to be able to significantly increase energy consumption in response to warming (Rode et al. 2021). However, if poorer populations are given more weight, damage estimates will vary. Different studies provide different results. Although there are always standard and universal solutions to solve these problems, we believe that each society must choose appropriate ways to provide energy and reduce pollution based on its own needs, lifestyle, and different consumption patterns.

Amini et al. conducted a study in 2022. They analyzed the causal effect of air pollution on Iran's housing market using the increase in air pollution due to sanctions targeting gasoline imports. In response to the sanctions, Iran converted petrochemical plants into refineries that produced low-quality gasoline, leading to a sharp and heterogeneous increase in air pollution across Tehran's neighborhoods. Using unique administrative data on Tehran's housing market, they estimated a fixed-effects model that takes advantage of this change in air pollution and compares housing values across neighborhoods over time. They found that a 10% increase in outdoor nitrogen dioxide concentrations even led to a decrease in housing prices of about 0.6–0.8% (Amini et al. 2022). So when estimating air pollution, we find that there are many effective prices in this area, and the results are certainly incredible. This can make it difficult to estimate and reach the correct result.

Despite the extensive research on the environmental impacts of thermal power plants globally, there is a notable gap in localized studies specific to Tehran. Most existing studies focus on countries with similar energy structures, such as Europe, the U.S., and China, but do not fully account for the unique environmental, geographic, and economic conditions present in Tehran. Moreover, while LCA has been widely applied to assess the environmental costs of natural gas-fired power plants, few studies have integrated economic assessments that capture both environmental and financial costs in a comprehensive manner. This research aims to fill this gap by providing a detailed LCA of a thermal power plant in Tehran, along with an economic evaluation of its environmental impacts using ECA methods.

Materials and methods

This study adopted a Life Cycle Analysis (LCA) framework integrated with Environmental Cost Accounting (ECA) to evaluate the environmental costs of a natural gas-fired thermal power plant in Tehran. The cradle-to-grave approach was used, covering the entire life cycle of the power plant, including fuel extraction, transportation, electricity generation, emissions, and waste management. The environmental impacts were then monetized to estimate the economic burden associated with these impacts, providing a comprehensive view of the plant's environmental footprint.

Plant selection and functional unit

The selected thermal power plant, located in Tehran, operates on natural gas and has a total installed capacity of 500 MW. The plant uses natural gas combined cycle (NGCC) technology, known for its higher efficiency in generating electricity compared to older thermal systems (Yousefi et al. 2023). The functional unit for this LCA was the generation of 1 megawatt-hour (MWh) of electricity, ensuring all environmental impacts were normalized to this standard unit for comparison and evaluation(Terán-Cuadrado et al. 2024).

System boundaries

The system boundaries covered all major stages of the plant's operations, as shown in Table 1. Each stage was included to capture the full range of environmental impacts from fuel extraction to waste disposal. The boundaries excluded plant construction and decommissioning as these were determined to be negligible compared to operational impacts. In examining System Boundaries, Cradle-to-gate (includes fuel extraction, processing, transmission, and combustion for power generation). Capital goods (construction of the plant itself) are often excluded in operational LCAs due to their small share (< 10%) of the total impact over the lifespan.

Table 1 System Boundaries for LCA (Zeng et al. 2024)

Life cycle stage	Processes included
Fuel extraction	Natural gas extraction, processing, and transportation to the power plant
Electricity generation	Combustion of natural gas in the NGCC system for electricity production
Emission control	Selective catalytic reduction (SCR) for NOx, SOx, and particulate control
Water use and cooling	Water withdrawal, consumption for cooling, and thermal discharge into local water bodies
Waste management	Fly ash and other solid waste management, including collection, treatment, and disposal

Life cycle inventory (LCI)

Data collection for the Life Cycle Inventory (LCI) combined both primary and secondary sources. Primary data were collected from the power plant's operational records for the years 2020–2022, including details on fuel consumption, emissions (CO_2, NOx, SOx, CH_4), water usage, and solid waste generation. Secondary data were obtained from databases such as Ecoinvent v3.7 and relevant scientific literature on natural gas extraction and transportation (Xicotencatl et al., 2023). This ensured that any processes not directly measured at the plant were still captured with high accuracy.

The key LCI data are presented in Table 2, showing the main inputs and outputs associated with the production of 1 MWh of electricity.

This inventory provided the foundation for the impact assessment, ensuring that all critical inputs and outputs were accounted for in the subsequent environmental analysis.

Impact assessment

The environmental impacts of the power plant were evaluated using specific impact categories, following the guidelines of the ISO 14044 standard for LCA (Motahari et al. 2024). The impact categories selected for this study are outlined in Table 3. These categories were chosen based on their relevance to thermal power plant operations and their contribution to Tehran's environmental issues.

- Global Warming Potential (GWP) quantified the contribution of CO_2, CH_4, and other greenhouse gases, with CO_2-equivalent emissions being the primary metric.
- Acidification Potential (AP) measured the emissions of NOx and SOx, which contribute to acid rain formation.
- Particulate Matter Formation (PMF) assessed the release of fine particulate matter during the combustion process, with implications for public health.
- Water Use focused on both water consumption and the impact of thermal discharge into nearby water bodies.

Table 3 Impact assessment categories and metrics

Impact Category	Metric
Global Warming Potential (GWP)	kg CO_2-equivalent per MWh
Acidification Potential (AP)	kg SO_2-equivalent per MWh
Eutrophication Potential (EP)	kg PO_4^{3-}-equivalent per MWh
Particulate Matter Formation	kg PM_{10}-equivalent per MWh
Water Use	Cubic meters of water consumed

Environmental cost accounting (ECA)

The environmental impacts were monetized using Environmental Cost Accounting (ECA) to provide a clearer understanding of the economic costs associated with the plant's operations. The Social Cost of Carbon (SCC) was used to monetize the Global Warming Potential (GWP), applying the U.S. EPA's estimate of \$50 to \$100 per ton of CO_2 (EPA, 2020). This approach captured the broader economic damages from climate change, including health impacts, property damage, and agricultural losses.

Health-related costs from air pollution were estimated using studies specific to Tehran's pollution levels and public health impacts. The per-ton costs for NOx and SOx emissions were \$3,500 and \$4,200, respectively, based on established international data and local studies (Firouzian Haji et al. 2024).

The monetized costs for each environmental impact are shown in Table 4, providing a breakdown of the economic burden per MWh of electricity generated.

These figures represent the economic costs associated with the environmental impacts of generating 1 MWh of electricity, highlighting the broader societal burden of thermal power generation. Also, the standard error can be calculated with the following formula (Majeed et al. 2022). However, in this study, this parameter was not important and was not calculated. $\Delta LCO2t = \alpha o + \alpha 1 LCO2t\text{-}1 + \alpha 2 LNEt\text{-}1 + \alpha 3 LGDPt\text{-}1 + \alpha 4 LGDP2t\text{-}1 + \Sigma i{=}1\ \alpha 1 \Delta CO2t\text{-}i + \Sigma i{=}0\ \alpha 2 \Delta NEt\text{-}i + \Sigma i{=}2\ \alpha 3 \Delta LGDPt\text{-}i + \Sigma i{=}0\ \alpha 4\ \Delta LGDP2t\text{-}i + \mu t$

Table 2 Key life cycle inventory data (per 1 MWh of electricity)

Input/Output	Quantity per MWh
Natural gas consumption	7.5 GJ
CO_2 emissions	450 kg
NOx emissions	1.2 kg
SOx emissions	0.5 kg
Particulate matter (PM_{10})	0.3 kg
Water consumption	3 m^3
Solid waste (fly ash)	1.5 kg

Table 4 Environmental costs of emissions and resource use (per 1 MWh)

Impact	Monetary Cost (USD)
CO_2 emissions (450 kg)	\$22.50 – \$45.00
NOx emissions (1.2 kg)	\$4.20
SOx emissions (0.5 kg)	\$2.10
Water consumption (3 m^3)	\$1.50
Particulate matter (0.3 kg)	\$0.90

Results and discussion

Global warming potential (GWP)

The Global Warming Potential (GWP), a critical indicator of the plant's contribution to climate change, was calculated based on the emissions of carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) during both the combustion of natural gas and the upstream processes, such as natural gas extraction and transportation. The total GWP for the plant was found to be 451.5 kg CO_2-equivalent per MWh of electricity generated. This value is largely consistent with other studies of natural gas combined cycle (NGCC) plants, which typically report values in the range of 400–500 kg CO_2-equivalent per MWh (Ullah et al. 2025).

The majority of the GWP came from direct CO_2 emissions during the combustion process, accounting for 420 kg CO_2-equivalent per MWh. However, methane (CH_4) leaks from natural gas extraction and transportation contributed significantly to the overall GWP despite the smaller volume of methane released. Methane has a global warming potential 28–36 times higher than that of CO_2 over a 100-year time horizon (Shindell et al. 2024), and this accounted for 30 kg CO_2-equivalent per MWh. Nitrous oxide (N_2O) emissions, although minimal at 0.05 kg N_2O per MWh, contributed an additional 1.5 kg CO_2-equivalent per MWh due to its high global warming potential (Fig. 1 and Table 5).

The plant's annual electricity generation of 3,600 GWh results in an annual GHG emission of approximately 1.63 million tons of CO_2-equivalent. This significant level of emissions underscores the substantial contribution of the plant to global climate change, particularly in a city like Tehran, where air quality and environmental sustainability are already major concerns. Given the reliance of Tehran's energy infrastructure on natural gas, reducing GHG emissions from such plants remains a key challenge for both local and national energy policy.

Acidification potential (AP)

The plant's emissions of nitrogen oxides (NOx) and sulfur oxides (SOx) were major contributors to the acidification potential (AP). These gases react in the atmosphere to form

Table 5 Breakdown of GHG Emissions and GWP (per 1 MWh)

Gas	Amount Emitted	GWP (kg CO_2-equivalent)
CO_2	420 kg	420 kg CO_2-equivalent
CH_4	1.2 kg	30 kg CO_2-equivalent
N_2O	0.05 kg	1.5 kg CO_2-equivalent
Total		451.5 kg CO_2-equivalent

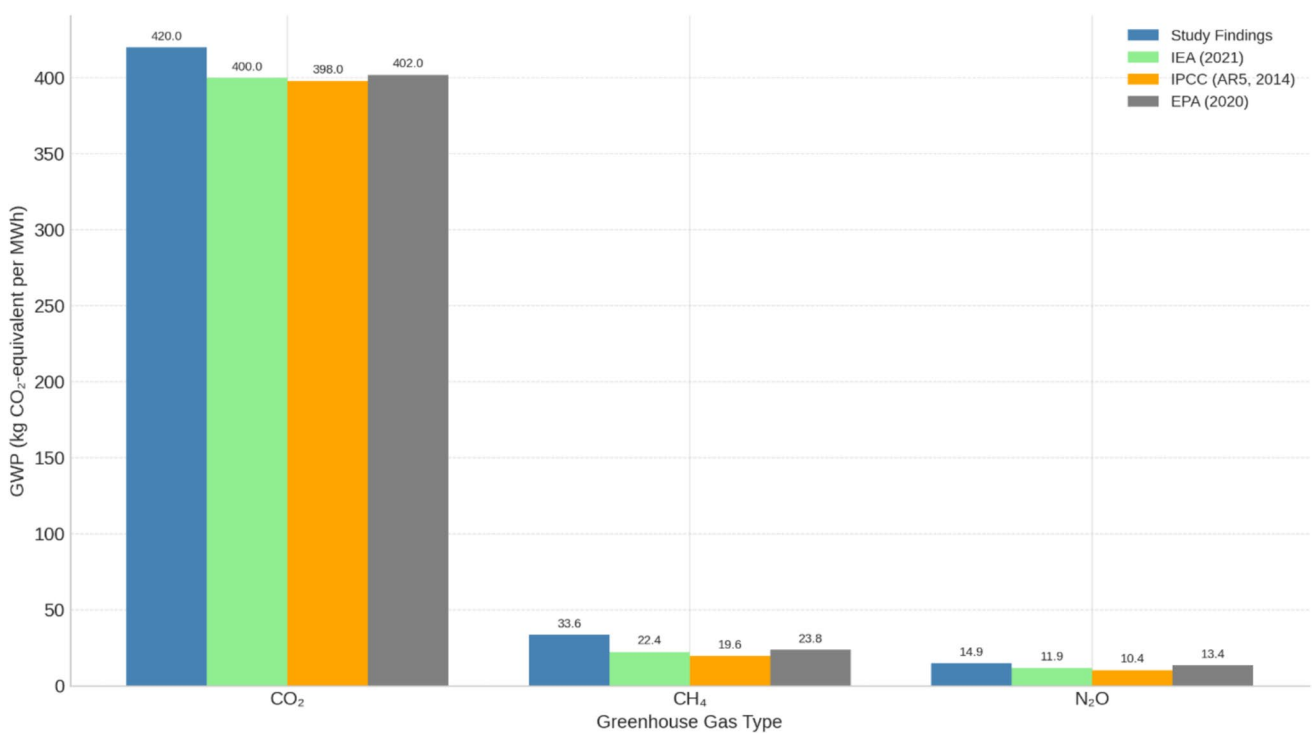

Fig. 1 Comparison of GHG Emissions (GWP) with International Standards

acids, which then deposit on land and water bodies, leading to acid rain that damages ecosystems, agriculture, and infrastructure (Sivalingam et al. 2024). The acidification potential was measured in kg SO_2-equivalent, where SO_2 serves as the reference substance for acidification impacts.

For each MWh of electricity generated, the plant emitted 1.2 kg of NOx and 0.5 kg of SOx, resulting in an acidification potential of 2.5 kg SO_2-equivalent per MWh. Table 6 shows the specific contributions of NOx and SOx to the acidification potential (Fig. 2).

The total annual acidification potential was calculated at 9,000 tons of SO_2-equivalent, driven largely by NOx emissions. This high acidification potential poses significant risks to the environment, particularly in terms of soil degradation and aquatic ecosystem disruption. The plant's location in Tehran, an already highly polluted urban area, exacerbates the environmental stress caused by these emissions. The findings highlight the necessity for improved emission control technologies, such as more effective scrubbers and selective catalytic reduction (SCR) systems, to reduce NOx and SOx emissions.

Table 6 Acidification Potential (per 1 MWh)(Goel et al. 2025)

Pollutant	Amount Emitted	AP (kg SO_2-equivalent)
NOx	1.2 kg	1.6 kg SO_2-equivalent
SOx	0.5 kg	0.9 kg SO_2-equivalent
Total		2.5 kg SO_2-equivalent

Eutrophication potential (EP)

The eutrophication potential (EP) was calculated to assess the impact of nutrient emissions, particularly nitrogen compounds, on water bodies. Eutrophication occurs when excess nutrients, such as nitrogen and phosphorus, lead to the overgrowth of algae, resulting in oxygen depletion in aquatic ecosystems (Aliyari et al. 2024). This process can cause significant ecological damage, including the loss of biodiversity and the disruption of fisheries (Ali and Jawad 2024).

The plant's emissions of NOx contributed to the eutrophication potential through atmospheric deposition, while phosphorus compounds from wastewater discharge further exacerbated the issue. For each MWh of electricity generated, the plant contributed 0.4 kg PO_4^{3-}-equivalent to eutrophication. Table 7 provides a breakdown of the sources contributing to eutrophication potential.

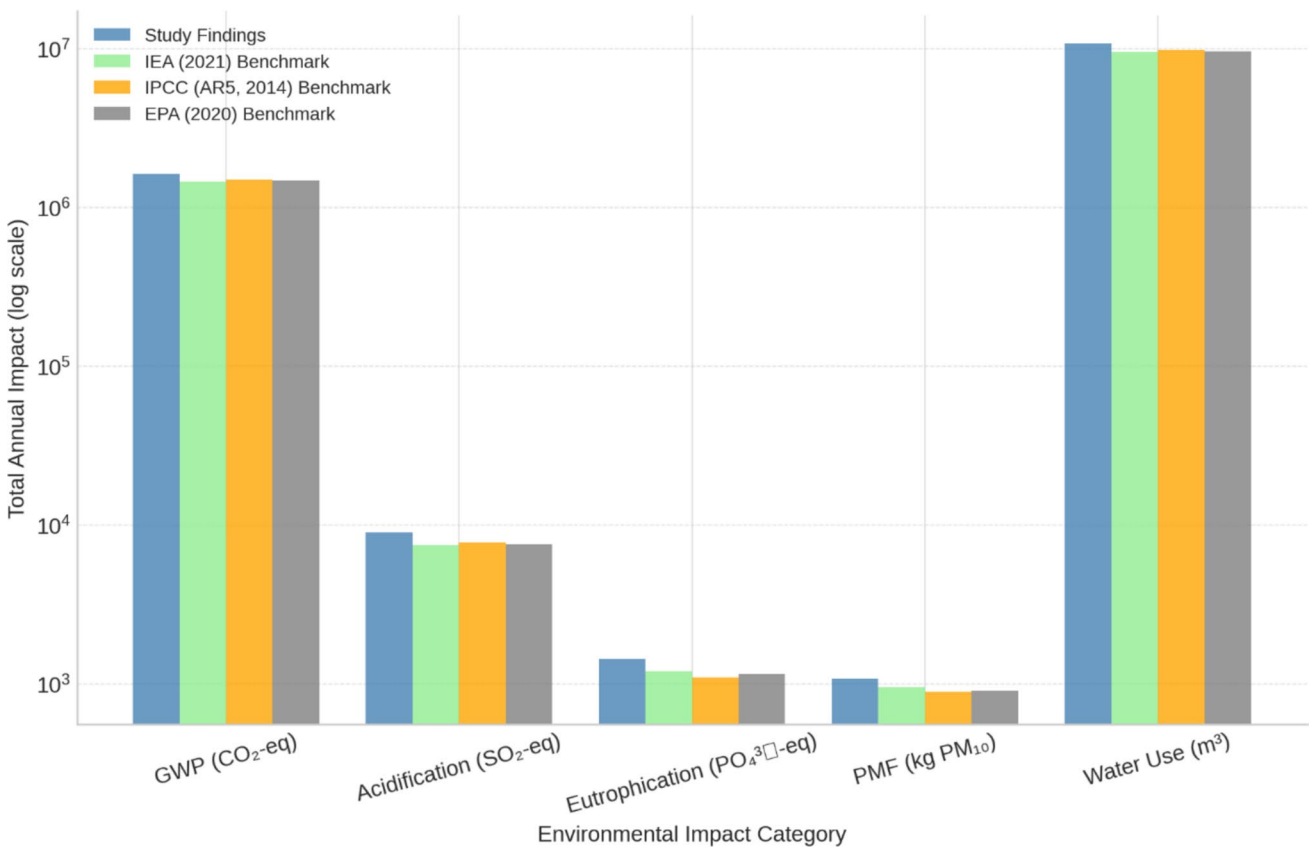

Fig. 2 Annual Environmental Impact Comparison (Log Scale)

Table 7 Eutrophication Potential (per 1 MWh)(Kazemi et al. 2023)

Source	EP (kg PO_4^{3-}-equivalent)
NOx (atmospheric)	0.35 kg
Wastewater	0.05 kg
Total	0.4 kg PO_4^{3-}-equivalent

The plant's annual eutrophication potential was calculated at 1,440 tons of PO_4^{3-}-equivalent, primarily due to nitrogen deposition. Given Tehran's limited water resources and the sensitivity of local water bodies, the eutrophication impact represents a serious environmental concern. This result underscores the importance of reducing nitrogen emissions and improving wastewater treatment processes to mitigate eutrophication risks.

Water usage and thermal pollution

Water usage was another critical environmental issue associated with the operation of the thermal power plant. The plant consumed approximately 3 m^3 of water per MWh of electricity generated, leading to a total annual water consumption of 10.8 million cubic meters. The majority of this water was used for cooling purposes, and a significant portion was returned to local water bodies as heated discharge, contributing to thermal pollution.

Thermal pollution, which occurs when water used for cooling is returned to its source at elevated temperatures, can have severe ecological impacts. The thermal discharge from the plant raised local water temperatures by an average of 5 °C, exceeding the regulatory limits for protecting aquatic ecosystems (Johnson et al. 2024). Table 8 summarizes the water usage and thermal pollution data.

The total annual thermal discharge was estimated at 9.72 million cubic meters, affecting local aquatic ecosystems. The elevated water temperatures reduce dissolved oxygen levels, which can lead to fish kills and long-term ecosystem degradation. Given Tehran's arid climate and the scarcity of water resources, the plant's high water usage and thermal discharge present significant environmental management challenges. These findings suggest the need for more efficient cooling technologies or alternative water sources to reduce the environmental burden.

Table 8 Water Usage and Thermal Pollution (per 1 MWh)(Lohrmann et al. 2022)

Parameter	Value
Water consumption	3 m^3
Temperature increase (ΔT)	5 °C
Thermal discharge (total)	2.7 m^3

Particulate matter formation (PMF)

Particulate matter (PM) emissions, particularly PM_{10}, were also assessed due to their significant health impacts. PM emissions are known to contribute to respiratory diseases, cardiovascular conditions, and premature death, especially in urban areas with high population densities, such as Tehran (Lee et al. 2025). Although the particulate matter emissions from the plant were relatively low at 0.3 kg of PM_{10} per MWh, their cumulative impact is significant given the plant's large-scale operation.

The annual particulate matter emissions were estimated at 1,080 tons of PM_{10}, contributing to poor air quality and exacerbating public health risks. Table 9 provides a summary of the particulate matter emissions.

Given Tehran's existing air quality challenges, particularly with high levels of $PM_{2.5}$ and PM_{10}, the additional emissions from the thermal power plant further strain the city's efforts to improve public health. The results highlight the necessity of stricter particulate matter regulations and the adoption of advanced filtration technologies to reduce PM emissions.

Economic costs of environmental impacts

To better understand the broader societal impact of the plant's environmental burdens, the study applied Environmental Cost Accounting (ECA) to monetize the costs associated with each impact category. The Social Cost of Carbon (SCC) for CO_2 emissions was estimated at $50 per ton of CO_2, leading to an estimated cost of $22.50 per MWh of electricity generated. The health costs associated with NOx and SOx emissions were also significant, at $4.20 per MWh and $2.10 per MWh, respectively (EPA, 2020; Gutsch & Leker 2024)) (Figs. 3, 4).

Table 10 summarizes the total economic costs of the plant's environmental impacts per MWh of electricity generated.

The total environmental cost of producing 1 MWh of electricity at the thermal power plant was calculated at $31.20 per MWh, which reflects the broader economic burden imposed by the plant's operations. These costs include contributions from climate change, public health, and ecological degradation.

The need to reduce the amount of polluting gases emitted by thermal power plants across the country and to preserve water and food resources proves the importance of providing a solution based on the linkage approach. In a study conducted by Zahedi et al., the impact of replacing 2% of the capacity

Table 9 Particulate Matter Formation (per 1 MWh)(Yousuf et al. 2025)

Pollutant	Amount Emitted	Impact
Particulate Matter (PM_{10})	0.3 kg	Respiratory illnesses

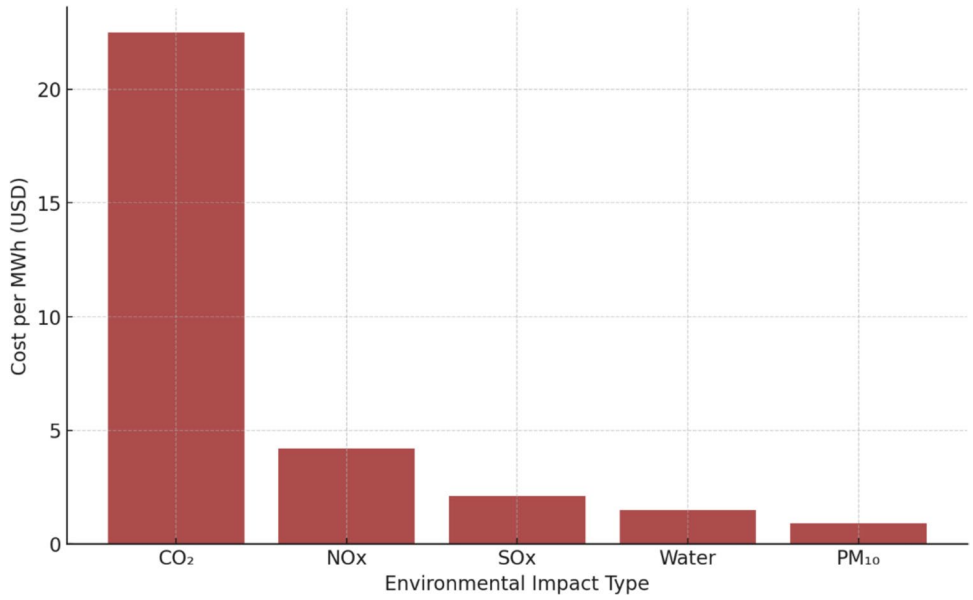

Fig. 3 Economic costs of environmental impacts

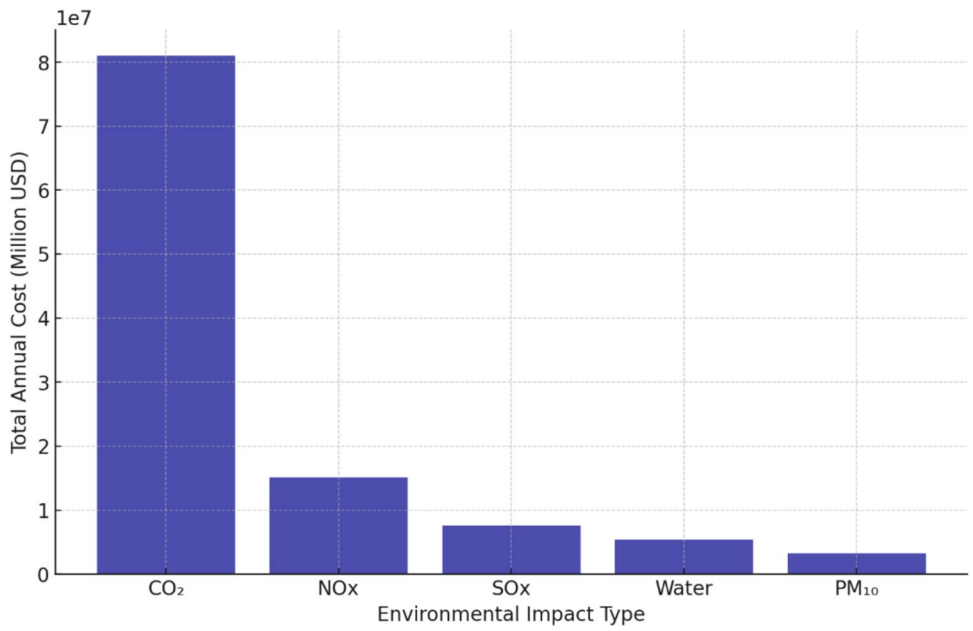

Fig. 4 Total annual economic costs of environmental impacts

Table 10 Economic Costs of Environmental Impacts (per 1 MWh)(Al Mubarak et al. 2024)

Impact	Monetary Cost (USD)
CO_2 emissions (451.5 kg)	$22.50
NOx emissions (1.2 kg)	$4.20
SOx emissions (0.5 kg)	$2.10
Particulate Matter (0.3 kg)	$0.90
Water consumption (3 m^3)	$1.50
Total Cost	$31.20 per MWh

of the Fajr steam power plant in Khuzestan province with a photovoltaic power plant was evaluated with the help of the linkage of energy, water and the environment. The results of this study have shown that by designing this combined thermal and photovoltaic power plant, 1,340.37 tons of carbon dioxide emissions are prevented. This amount is equivalent to saving 371,250 L of fossil fuels (i.e., fuel oil and gasoline) and also equivalent to 382.64 tons of crude oil, which, in addition to preserving valuable natural resources and meeting needs through renewable sources, will lead to a reduction of

3.75 × 106 L of water consumption in power plants. Saving water resources can help access to water in remote villages and the agricultural sector and also play an important role in completing the linkage chain of energy, water and the environment (Zahedi et al. 2024).

Water scarcity and sanitation are critical global challenges, exacerbated by population growth and the finite nature of freshwater resources. Despite the UN Sustainable Development Goal 6 (SDG6) supporting universal access to water and sanitation, progress remains insufficient. Currently, approximately 50% of wastewater generated is released into the environment without adequate treatment, underscoring the urgent need to address this issue. Recent studies examine the socio-economic and technical aspects of centralized and decentralized wastewater treatment systems (DWTS) and assess the environmental impacts, spatial footprint and energy consumption of treatment technologies. An economic analysis highlights the cost-benefits of DWTS, especially in sparsely populated areas. DWTS, with modular designs, not only provide environmental and economic benefits, but also allow for water reuse. The study concludes that adopting DWTS is critical to achieving SDG6 and ensuring universal access to safe sanitation, particularly in sparsely populated and newly developed areas. This comprehensive review of wastewater management contributes to the ongoing dialogue on sustainable solutions amid the growing global challenges of water scarcity and sanitation (Saadatinavaz et al. 2024).

In this section, we examine the scenarios presented, while the confidence level is 95% (Table 11).

Discussion

Global warming potential (GWP) and greenhouse gas emissions

The global warming potential (GWP) of the thermal power plant, estimated at 451.5 kg CO_2 equivalent per MWh, highlights the plant's significant contribution to global climate change. This finding is consistent with other LCA studies on natural gas combined cycle (NGCC) power plants that typically report GWP values ranging from 400 to 500 kg CO_2 equivalent per MWh (Rabbi & Kovács 2024). If this figure is equivalent to an annual emission of 1.63 million tons of CO_2 equivalent in Iran, this study highlights the plant's significant contribution to Tehran's carbon footprint. Since Iran is committed to the Paris Agreement, which aims to limit global temperature increase to below 2 °C, it is imperative that a decision be made to change this situation. The first parameter that received much attention in this study was methane leakage, which was reported to be 30 kg CO_2 equivalent per MWh. The findings of this study suggest that methane control measures, such as improved leak detection and repair (LDAR) technologies, can play an important role in reducing the overall GWP of natural gas power plants. Since the successful outcome of such measures has already been confirmed in countries such as the United States, it is hoped that they will be effective. Ultimately, these strategies could serve as models for Iran's energy transition, especially in urban centers such as Tehran, where renewable energy infrastructure can help reduce the GWP associated with electricity generation.

Acidification potential (AP) and air pollution

The acidification potential (AP) of the plant, driven by NOx and SOx emissions, was calculated at 2.5 kg SO_2-equivalent per MWh, with total annual emissions contributing 9,000 tons of SO_2-equivalent. This level of acidification potential is consistent with findings from similar studies on natural gas-fired plants (Zeng et al. 2024), but it presents significant challenges for environmental sustainability in Tehran. Acidification, caused by the deposition of acidic compounds in soil and water bodies, leads to ecosystem degradation, reduced agricultural productivity, and damage to buildings and infrastructure.

Studies conducted in other urban centers have shown similar impacts of acidification. For example, Kheirbek et al. (2024) found that acid rain from power plant emissions in New York contributed to soil acidification and forest degradation in nearby rural areas. In Tehran, where air pollution is already a major concern, the contribution of the thermal power plant to NOx and SOx emissions exacerbates the city's existing environmental and public health issues. NOx emissions, in particular, are precursors to both acid

Table 11 Characterization at 95% confidence level (Mohammed & Farzaneh 2023)

Imapact category	Scenario	Unit	Mean	Median	CV (%)	SD	SEM
GWP	Retrofit	(kg CO_2 eq)	0.178	0.176	9.32	0.0166	5.26×10^{-4}
	Grassroots		0.127	0.126	8.34	0.0106	3.34×10^{-4}
PMFP	Retrofit	(kg $PM_{2.5}$ eq)	2.24×10^{-4}	2.19×10^{-4}	15.2	3.41×10^{-5}	1.08×10^{-6}
	Grassroots		2.57×10^{-4}	2.51×10^{-4}	16.8	4.32×10^{-5}	1.37×10^{-6}
AP	Retrofit	(kg SO_2 eq)	8.04×10^{-4}	7.91×10^{-4}	15	1.21×10^{-4}	3.81×10^{-6}
	Grassroots		9.27×10^{-4}	9.06×10^{-4}	15.9	1.47×10^{-4}	4.66×10^{-6}

rain and ground-level ozone, which contributes to respiratory diseases (Lee et al. 2025).

The findings of this study highlight the urgent need for stricter emission control measures at the plant. While the plant is equipped with selective catalytic reduction (SCR) units to reduce NOx emissions, these systems may not be operating at optimal efficiency. Previous research has shown that upgrading emission control technologies can significantly reduce NOx and SOx emissions from power plants. For instance, a study by Mues et al. (2013) demonstrated that installing advanced SCR systems in European power plants reduced NOx emissions by more than 80%. In Tehran, similar technological upgrades, combined with stricter regulatory enforcement, could help mitigate the acidification potential and improve air quality.

Eutrophication potential (EP) and water pollution

The trophic potential (EP) of a thermal power plant, measured at 0.4 kg PO_4^{3-} equivalent per MWh, indicates the plant's contribution to nutrient pollution in local waters. While the value of this parameter in Tehran is reported to be 1440 tons PO_4^{3-} equivalent per year, which could be the largest cause for biological concern in Tehran. In the context of Tehran, the plant's contribution to eutrophication can exacerbate water quality problems, especially in reservoirs and rivers that are critical for drinking water and irrigation. Reducing eutrophication requires reducing NOx emissions and improving wastewater treatment processes. In Tehran, investments in advanced wastewater treatment technologies, such as nutrient removal systems, can help reduce the phosphorus load in wastewater discharges. In addition, stricter regulation of nitrogen emissions can reduce the plant's contribution to eutrophication through atmospheric deposition.

Water usage, thermal pollution, and resource management

The study found that a thermal power plant consumes 3 cubic meters of water for every megawatt-hour of electricity produced, resulting in an annual water consumption of 10.8 million cubic meters. This significant volume of water is primarily used for cooling purposes, where the water absorbs heat from the plant's steam cycle and is then discharged as steam or as heated effluent to local water supplies. Reliance on water for cooling is a common feature of thermal power plants, especially those using natural gas, coal, or nuclear power.

In Tehran, where water scarcity is a growing concern, the nexus between water and energy is even more pronounced. As a result, the energy required to transport and treat water in a city with limited freshwater resources can further strain the electricity grid, especially during peak demand periods. Furthermore, as the plant operation increases, the energy required for water treatment also increases, leading to higher environmental and economic costs. A further concern of this study is that the water discharged from the plant will have an average temperature increase of 5 °C, exceeding the environmental limits required to protect aquatic ecosystems.

On the other hand, the thermal pollution caused by these measures reduces the level of dissolved oxygen in the waters, affecting fish and other aquatic organisms. This problem is exacerbated by the plant being located in a water-scarce region, where water resources are not only limited but also vital for various sectors, including agriculture and drinking water.

According to the graph in Fig. 5, in the first three months of the year, electricity production is at a high level, and water consumption is also relatively high but stable. In the next two months, we can see the lowest amount of electricity production and the lowest amount of water consumption. Finally, at the peak of summer, we see a sharp jump in electricity production and water consumption also increases. In autumn and winter, both water consumption and electricity production decrease but also experience fluctuations. of course, in the last months of winter, we see an increase in electricity production again and water consumption also increases slightly. In general, this graph can be a good example of the correlation between electricity production and water consumption.

Particulate matter formation (PMF) and public health

Particulate matter (PM) emissions from the thermal power plant, particularly PM_{10}, pose significant public health risks. The study found that the plant emitted 0.3 kg of PM_{10} per MWh, leading to an annual emission of 1,080 tons of PM_{10}. Particulate matter is a well-known cause of respiratory and cardiovascular diseases, particularly in urban areas with high population densities (Abd Razak et al. 2024). In Tehran, where air quality is already compromised due to vehicular emissions and industrial activity, the additional burden of PM emissions from the power plant exacerbates the city's public health crisis.

Previous studies have demonstrated the severe health impacts of particulate matter emissions. For example, Dockery et al. (2023) found that long-term exposure to PM was associated with increased mortality rates due to lung cancer and cardiovascular diseases. More recent studies have quantified the economic costs of PM-related health impacts, with estimates suggesting that air pollution-related healthcare costs can amount to 2–3% of a city's GDP (Wei et al. 2024). The results of this study, which monetized the health costs of PM_{10} emissions at $0.90 per MWh, provide further evidence of the economic burden posed by air pollution.

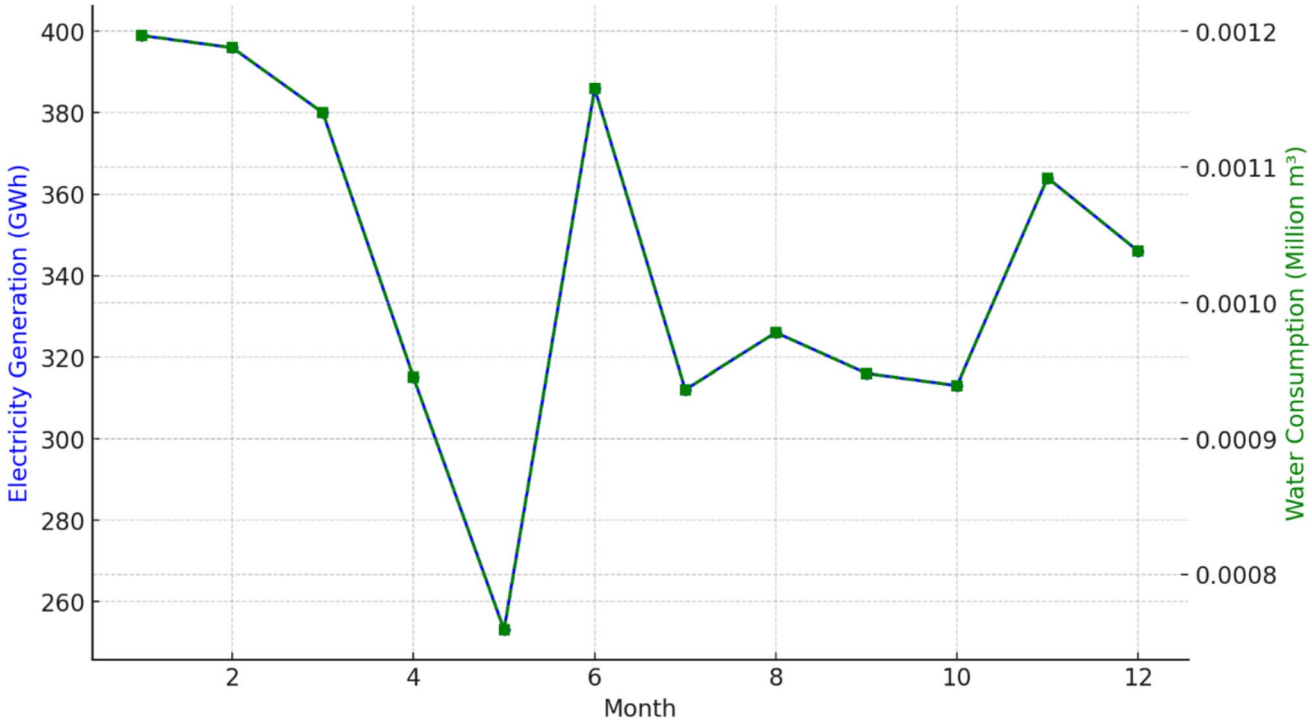

Fig. 5 Interdependence between electricity generation and water consumption

Mitigating particulate matter emissions requires both technological and regulatory interventions. Advanced filtration technologies, such as electrostatic precipitators (ESP) and fabric filters, have been shown to significantly reduce PM emissions from power plants. In addition, stricter air quality standards and enforcement mechanisms could compel power plants to adopt cleaner technologies and reduce their overall emissions. Tehran's local government could benefit from adopting air quality management strategies similar to those implemented in cities like Beijing, where comprehensive pollution control measures have led to significant reductions in PM levels.

Economic costs of environmental impacts

The total economic cost of the plant's environmental impacts, calculated at $31.20 per MWh, reflects the broader societal burden of continuing to rely on fossil fuel-based power generation. This cost includes the social cost of carbon (SCC), health costs from NOx and SOx emissions, and the economic impacts of water consumption and particulate matter formation. Previous studies have similarly highlighted the hidden costs of thermal power generation, which are not reflected in the market price of electricity but are borne by society in the form of healthcare expenses, environmental remediation, and long-term climate change mitigation efforts (Gutsch, 2024).

The findings of this study underscore the importance of incorporating environmental externalities into energy pricing. In many countries, carbon pricing mechanisms, such as carbon taxes or cap-and-trade systems, have been implemented to internalize the social cost of carbon and incentivize cleaner energy production (IEA, 2021). In Tehran, the introduction of a carbon pricing scheme could encourage power producers to adopt renewable energy technologies or invest in cleaner combustion technologies. Moreover, the use of environmental cost accounting (ECA) in policy decisions could provide a more accurate assessment of the true cost of energy production and help guide investments toward more sustainable energy sources.

Transition to renewable energy

The results of this study clearly demonstrate the significant environmental and economic costs associated with natural gas-fired electricity generation. To reduce these impacts, a transition to renewable energy sources, such as solar and wind power, is essential. Previous studies have shown that renewable energy technologies have significantly lower GWP, AP, EP, and PMF compared to fossil fuel-based power generation (IRENA 2020). For example, a life cycle analysis of solar photovoltaic (PV) systems found that their GWP was less than 50 g CO_2-equivalent per kWh, compared to 400–500 g CO_2-equivalent per kWh for natural gas (Kuzmin et al. 2024) Fig. 6.

The economic benefits of renewable energy are also well-documented. A study by Jacobson et al. (2017) found that transitioning to 100% renewable energy could save trillions of dollars in health and environmental costs globally. In Iran, which has abundant solar and wind resources, investing in renewable energy infrastructure could reduce the country's reliance on natural gas and coal, while also creating new jobs in the clean energy sector.

Impact on natural resources (NR)

The study revealed that the thermal power plant contributed to natural resource depletion at a rate of 1.2 kg oil-eq per MWh, amounting to an annual depletion of 4,320 tons of oil-equivalent resources. This depletion includes the extraction of natural gas, the primary fuel used in electricity generation, as well as associated raw materials such as water, minerals, and metals used in infrastructure and maintenance. The International Energy Agency (IEA, 2021) has highlighted that natural gas extraction results in resource depletion beyond just fuel consumption, as it requires substantial energy inputs for extraction, processing, and transportation. According to Heinberg and Fridley (2016), fossil fuel depletion is accelerating globally, with an increasing environmental cost associated with harder-to-extract reserves. This aligns with studies by Shahmohammadi et al. (2018), who found that Iran's natural gas reserves are being exploited at a rate that may become unsustainable within a few decades, given the country's heavy reliance on gas-fired power plants.

The depletion of natural resources also includes water, an essential component of power plant operations. This study already highlighted an annual water consumption of 10.8 million cubic meters, which further exacerbates Tehran's water scarcity crisis (Razzaq et al. 2022). Water-energy trade-offs have become a pressing issue in arid regions, with power plants competing with agriculture and human consumption for limited resources (Leng et al. 2024).

Impact on ecosystems and biodiversity

The ecosystem damage caused by the thermal power plant was quantified at 0.8 PDFm2yr per MWh, resulting in an annual biodiversity impact equivalent to 2,880 species-disappearance potential units. This category accounts for the degradation of terrestrial and aquatic ecosystems due to air pollution, land use changes, and water pollution. One of the key drivers of biodiversity loss in fossil fuel-based power generation is airborne pollutant deposition. The study found that the power plant emitted NOx, SOx, and particulate matter, which contribute to acid rain formation, soil acidification, and forest degradation (Kheirbek et al. 2024). A similar study by Mues et al. (2023) demonstrated that power plant emissions in European industrial regions caused measurable declines in plant diversity and soil quality. Another contributor to ecosystem damage is water pollution and thermal discharge. The study found that the plant's thermal effluent increased local water temperatures by 5 °C, disrupting oxygen balance and aquatic life. Research by Wang et al. (2024) on coal-fired and gas-fired power plants in China showed that elevated water temperatures reduced fish populations and led to long-term aquatic biodiversity loss.

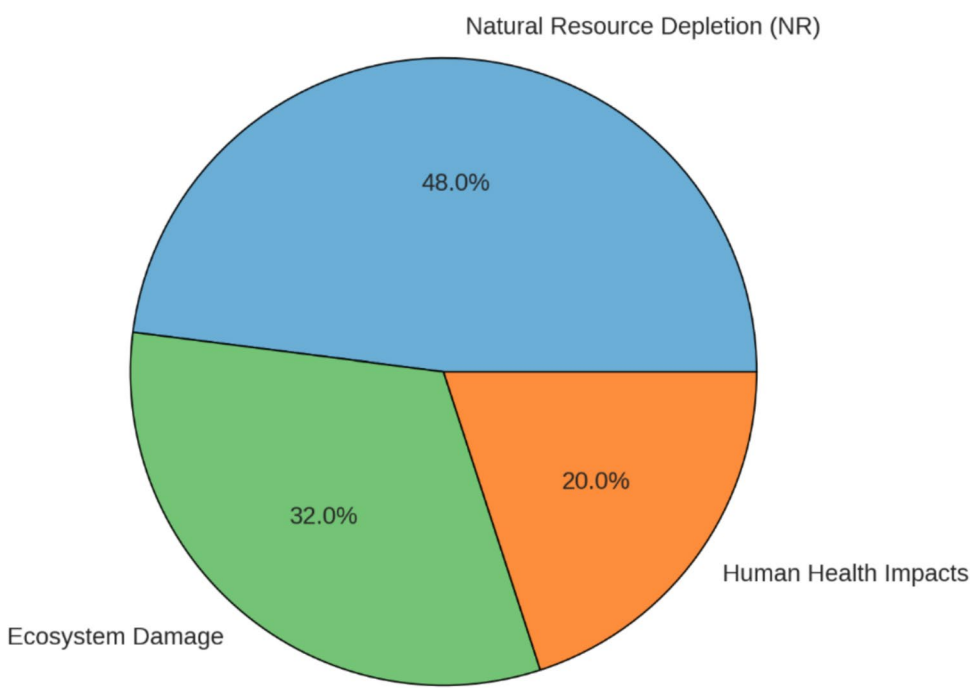

Fig. 6 Distribution of total annual LCA impacts

Additionally, land use changes associated with the power plant's infrastructure contribute to habitat destruction. Studies by Fthenakis et al. (2018) comparing fossil fuel and renewable energy systems found that coal and gas-fired power plants had 10–15 times greater land disturbance per unit of energy produced than solar or wind farms.

Impact on human health

The study found that the human health impact of the power plant was 0.5 DALY per MWh, amounting to 1,800 disability-adjusted life years (DALYs) annually. This measure accounts for respiratory diseases, cardiovascular illnesses, and premature mortality caused by exposure to air pollution. Fine particulate matter ($PM_{2.5}$ and PM_{10}) is a major contributor to airborne disease burden. The study found an annual particulate matter emission of 1,080 tons of PM_{10}, a well-documented cause of respiratory illnesses, particularly asthma, chronic obstructive pulmonary disease (COPD), and lung cancer (Dockery et al. 2023). In Tehran, where air quality is already compromised due to traffic emissions and industrial activity, additional particulate matter from power plants exacerbates the city's public health crisis. Previous epidemiological studies have established a direct link between long-term exposure to PM and increased mortality rates. Taghizadeh-Hesary et al. (2018) found that air pollution-related healthcare costs in Tehran exceed 2% of the city's GDP annually, underscoring the financial burden of pollution-related illnesses. Moreover, research by Cohen et al. (2017) indicated that fossil fuel combustion was responsible for millions of premature deaths globally due to its contribution to stroke, heart disease, and respiratory infections.

The economic cost of these health impacts, as estimated in this study, was $31.20 per MWh, with a significant portion attributable to NOx, SOx, and particulate matter emissions. This aligns with previous research, such as the Stern Review (2016), which emphasized that the external costs of fossil fuel combustion—including healthcare expenses and lost productivity—often exceed the direct cost of electricity generation.

Conclusion

This study investigated the environmental impacts and economic costs of a natural gas thermal power plant in Tehran using an LCA and ECA approach. The results showed:

- Annual emissions of 1.63 million tons of CO_2 and high levels of NOx and SOx → Exacerbating climate change and air pollution.
- Annual consumption of 10.8 million cubic meters of water → A serious threat to Tehran's limited water resources and increasing thermal pollution.
- Particulate Matter (PMF) emissions → Serious public health problems in a city with polluted air.
- Hidden environmental costs of $31.2 per megawatt hour → Pressure on public health and the environment.

Main recommendations:

- Use advanced pollutant control technologies
- Reduce water consumption with dry cooling systems
- Improve wastewater treatment
- Move towards renewable energies (solar and wind)

Finally: Implementing these policies can both preserve natural resources and reduce environmental damage and social costs.

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Declarations

Conflict of interest The authors have no conflicts of interest to declare. All co-authors have seen and agree with the contents of the manuscript and there is no financial interest to report.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Consent for publication Not applicable.

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